

Supervised Learning Capstone: Ames Housing Price Prediction

Machine Learning Track

Problem Statement

You are tasked with building a robust supervised learning pipeline to predict real estate prices based on a mix of categorical and numerical features. You will apply the full suite of algorithms you have learned, from foundational linear models to tree-based ensembles.

Dataset Link: Kaggle: Ames Housing Dataset

Available Features: Id, MSSubClass, MSZoning, LotFrontage, LotArea, Street, Alley, LotShape, LandContour, Utilities, LotConfig, LandSlope, Neighborhood, Condition1, Condition2, BldgType, HouseStyle, OverallQual, OverallCond, YearBuilt, YearRemodAdd, RoofStyle, RoofMatl, Exterior1st, Exterior2nd, MasVnrType, MasVnrArea, ExterQual, ExterCond, Foundation, BsmtQual, BsmtCond, BsmtExposure, BsmtFinType1, BsmtFinSF1, BsmtFinType2, BsmtFinSF2, BsmtUnfSF, TotalBsmtSF, Heating, HeatingQC, CentralAir, Electrical, 1stFlrSF, 2ndFlrSF, LowQualFinSF, GrLivArea, BsmtFullBath, BsmtHalfBath, FullBath, HalfBath, BedroomAbvGr, KitchenAbvGr, KitchenQual, TotRmsAbvGrd, Functional, Fireplaces, FireplaceQu, GarageType, GarageYrBlt, GarageFinish, GarageCars, GarageArea, GarageQual, GarageCond, PavedDrive, WoodDeckSF, OpenPorchSF, EnclosedPorch, 3SsnPorch, ScreenPorch, PoolArea, PoolQC, Fence, MiscFeature, MiscVal, MoSold, YrSold, SaleType, SaleCondition.

Project Checkpoints

Checkpoint 1: Overall Flow Presentation

Deliverable: A brief presentation (or diagram) of your planned machine learning pipeline.

- Define the target variable for Regression (**SalePrice**).
- Define a derived target variable for Classification (e.g., binning **SalePrice** into "Budget", "Standard", "Premium").
- Outline the validation strategy (e.g., K-Fold Cross-Validation).

Checkpoint 2: Mid-Project Review (EDA & Cleaning)

Deliverable: Cleaned dataset and exploratory data analysis report.

- **Missing Values:** Demonstrate how you handled NaN values (Note: NaN in **Alley** or **PoolQC** means "None", not missing data).
- **Scaling & Encoding:** Show your application of **StandardScaler**/**MinMaxScaler** and **One-Hot Encoding**.
- **Feature Engineering:** Create at least two new features (e.g., Total Square Footage, House Age).

Checkpoint 3: Final Delivery & Benchmarking

Deliverable: Final code, visualizations, and the completed benchmarking table.

- **Hyperparameter Tuning curves:** For all models, plot the **Training vs. Validation Loss** across different hyperparameter values (e.g., varying SVM's `C` parameter or Random Forest's `max_depth`) to demonstrate where overfitting/underfitting occurs.
- Use the model you trained to fill the test.csv and submit it.
- **Performance Table:** Fill out the following table using your test set or cross-validation scores.

Model	Regression Metric (RMSE)	Classification Metric (F1-Score)
Linear / Logistic Regression		
Polynomial Regression		
SVM (Linear Kernel)		
SVM (RBF Kernel)		
Decision Tree		
Random Forest		

Table 1: Final Model Benchmarking Results